

Assignment 1 Q1

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```
library(dplyr) library(ggplot2)
```

Q1

(a)

```
total_distance_2d = function(x, y) {  
  dx = diff(x)  
  dy = diff(y)  
  distances = sqrt(dx^2 + dy^2)  
  return(sum(distances))  
}
```

(b)

```
mean_abs_change_in_direction = function(x, y) {  
  dx = diff(x)  
  dy = diff(y)  
  
  directions = atan2(dy, dx)  
  dtheta = diff(directions)  
  
  mean_abs_change = mean(abs(dtheta))  
  
  return(mean_abs_change)  
}
```

(c)

```
max_abs_curvature_2d = function(x, y) {  
  
  dx = diff(x)  
  dy = diff(y)
```

```

    ddx = diff(dx)
    ddy = diff(dy)

    dx_adj = dx[-length(dx)]
    dy_adj = dy[-length(dy)]

    numerator = dx_adj * ddy - dy_adj * ddx
    denominator = (dx_adj^2 + dy_adj^2)^(3/2)

    curvature = numerator / denominator

    return(max(abs(curvature)))
}

```

(d)

```

p80_percent_acceleration = function(x, y){

    dx = diff(x)
    dy = diff(y)

    ddx = diff(dx)
    ddy = diff(dy)

    acc = sqrt(ddx^2 + ddy^2)

    acc_ordered = sort(acc)
    index = round(length(acc_ordered) * 0.8)

    if (index == 0) index = 1

    return(acc_ordered[index])
}

```

(e)

##	test_cases	Bezier	Space	Heart
## 1	Mean Velocity	0.12210	1.013e+00	1.0210
## 2	Total Distance	12.21000	2.442e+02	102.1000
## 3	Mean Abs Direction Change	0.01821	3.317e-01	0.1291
## 4	Max Abs Curvature	0.21210	3.940e+07	4.1970
## 5	80th Percentile Acceleration	0.00277	2.652e-01	0.1720